

Assignment 1

1. Perform the following base conversions. [Marks: 5]
 - I. $(100010010101)_{2s} = (?)_{12}$
 - II. $(001000101)_{1s} = (?)_7$
 - III. $(10110111)_{\text{Excess-5}} = (?)_5$
 - IV. $(01011010100011)_{2s} = (?)_{10}$
 - V. $(8127532.48)_9 = (?)_3$
2. Suppose you have two 6-bit registers where your computer stores signed numbers. Now subtract 27 from -20 using your computer's registers, and figure out whether there is any overflow or not. Use 1's complement to perform the computations. [Marks: 5]
3. A computer uses 10-bit registers to store numbers. [Marks: 4]
 - I. Calculate the lowest and highest possible unsigned numbers.
 - II. Calculate the lowest and highest possible numbers in the Sign & Magnitude system.
 - III. Calculate the lowest and highest possible numbers in the 1's complement number system.
 - IV. Calculate the lowest and highest possible numbers in the 2's complement number system.
4. Divide $(421260587)_9$ by $(260)_9$ and find the quotient and remainder if any. [Marks: 5]
5. Bangladesh is playing against India in a T20 cricket match. India scored $(190)_{11}$ runs in $(20)_{10}$ overs. Then Bangladesh scored $(141)_{13}$ runs.
 - I. Calculate Bangladesh's run rate in base-11 (show up to 5 decimal places if needed). [Marks: 6]
 - II. Identify which team won the match and the difference between the two teams' runs in base-8. [Marks: 5]